The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 Mar 94.

INCH-POUND

MIL-S-19500/359D
10 December 1993
SUPERSEDING
MIL-S-19500/359C
27 February 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, FAST RECOVERY, POWER RECTIFIER,
1N4942, 1N4944, 1N4946, 1N4947, AND 1N4948
JAN, JANTX, AND JANTXV

Inactive for new design after 27 February 1992 for For new design use - 1N5615, 1N5617, 1N5619, 1N5621, 1N5623 on MIL-S-19500/429.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a silicon, fast recovery semiconductor power rectifier diode for use in equipment circuits. Three levels of product assurance are provided for each device as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1. (Similar to DO-41)

1.3 Maximum ratings.

Device type	V_{RWM} V_R	I_O 1/ 2/ T_A = +55°C	I_O 2/ T_A = +100°C	I_{FSM} at t_p = 8.3 ms T_A = +100°C	Barometric pressure	t_{rr1} at I_F = 0.5 A I_R = 1.0 A I_{REC} = 0.25 A	T_{STG} and T_{OP}
	V pk V dc	A dc	A dc	A pk	mm Hg	ns	°C
1N4942	200	1.0	.750	15		150	-65 to +175
1N4944	400	1.0	.750	15	8	150	-65 to +175
1N4946	600	1.0	.750	15	8	250	-65 to +175
1N4947	800	1.0	.750	15	33	250	-65 to +175
1N4948	1000	1.0	.750	15	33	500	-65 to +175

1/ Derate linearly from 1.0 A at T_A = +55°C to 0.75 A at +100°C. Derate linearly from 0.75 A to 0 A between +100°C and +175°C.
2/ No forced air nor heat sinking shall be permitted for I_O ratings at T_A = +55°C and +100°C.

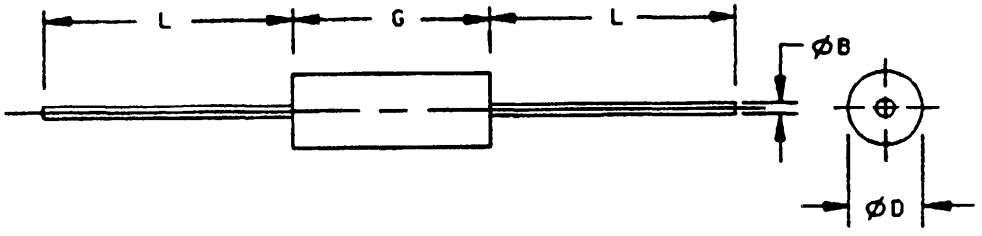
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
ØB	.027	.033	0.69	0.84	
ØD	.065	.150	1.65	3.81	3
G	.140	.250	3.56	6.35	3
L	1.000	1.500	25.40	38.10	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension G and ØD include all components of the diode periphery except the section of leads over which the diameter is controlled.
4. Dimension ØD shall be measured at the largest diameter.

FIGURE 1. Physical dimensions.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

T_{CVF} - - - - - - - - - - Temperature coefficient of forward voltage

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 herein. All devices shall be metallurgically bonded-thermally-matched-noncavity-double plug construction as defined in MIL-S-19500.

3.3.1 Encapsulant material. In addition to those categories of hermetically sealed package requirements specified in MIL-S-19500, fused metal oxide-to-metal shall also be acceptable.

3.3.2 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500, and as specified herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.5 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

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4.2 Screening (JANTX and JANTXV levels). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement JANTX and JANTXV levels
9	Not applicable
11	I_{R1} and V_{FM1}
12	See 4.2.1 and 4.5.3
13	Subgroup 2 of table I herein: ΔI_{R1} ≤ 100 percent of initial reading or ±50 nA dc whichever is greater. ΔV_{FM1} ≤ +0.1 V dc, -0.2 V dc Scope display evaluation (see 4.5.4).

4.2.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = room ambient as defined in the general requirements of paragraph 4.5 of MIL-STD-750; $V_R = V_{RWM}$ rated (see 1.3); F = 50-60 Hz; I_O = 1.0 A dc.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500 and table II herein.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein (see subgroup conditions for delta limits when applicable).

4.4.2.1 Group B inspection, table IVb (JAN, JANTX, and JANTXV of MIL-S-19500).

Subgroup	Method	Condition
B2	4066	I_{FSM} = rated (see 1.3); 10 surges of 8.3 ms each at 1-minute intervals; I_O = 0.75 A dc; V_{RSM} = 0.
B3	1027	I_O = 1.0 A dc minimum; f = 50-60 Hz; T_A = room ambient as defined in the general requirements of paragraph 4.5 of MIL-STD-750. V_R = rated V_{RWM} (see 1.3, 4.5.3, and 4.5.3.1); t = 340 hours. Delta limits: ΔI_R = 100 percent or 50 nA, whichever is greater.
B5	3101 or 4081	$R_{\theta JL}$ = 38°C/W; L = .375 inch (9.52)

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein (see subgroup conditions for delta limits when applicable).

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
C2	1056	Test Condition A
C2	2036	Tension: Test Condition A, weight = 5 pounds; t = 15 seconds.
C2	1021	Omit initial conditioning Delta limits: ΔV_F = -0.2 minimum and +0.1 maximum; ΔI_R = 100 percent or 50 nA, whichever is greater.
C5		Not applicable
C6	1026	T_A = room ambient as defined in the general requirements of paragraph 4.5 of MIL-STD-750; f = 50-60 Hz; I_O = 1.0 A minimum; V_R = rated V_{RWM} (see 1.3, 4.5.3, and 4.5.3.1)

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-S-19500.

4.4.4.1 Group E inspection, table VII of MIL-S-19500.

Subgroup	Method	Condition
E5	1001	1N4944, 1n4946 = 8 mm Hg (100,000 ft) 1N4947, 1n4948 = 33 mm Hg (70,000 ft) Voltage during test = V_{RWM}; maximum leakage (I_R) during test shall be 5 μA.

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature T_A of +25°C ±3°C.

4.5.3 Burn-in and life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees, nor less than 150 degrees.

4.5.3.1 Alternate mounting conditions. At the option of the manufacturer, any clips or heat sink mounting configurations may be utilized provided that I_O is increased such that the junction temperature of each diode is maintained at +145°C minimum.

4.5.4 Scope display evaluation. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 10 to 50 μA/division and 50 to 200 V/division. Reverse current over the knee shall be at least 10 μA and less than 50 μA peak. Any discontinuity below actual breakdown or dynamic instability of the trace shall be cause for rejection of that device.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical	2071					
Subgroup 2						
Forward voltage	4011	I_{FM1} = 1.0 A dc pulsed (see 4.5.1)	V_{FM1}	0.6	1.3	V dc
Reverse current leakage	4016	V_R = rated V_{RWM} (see 1.3) pulsed (see 4.5.1)	I_{R1}		1.0	µA
Breakdown voltage	4021	I_R = 50 µA pulsed (see 4.5.1)	$V_{(BR)1}$			
1N4942				220		V dc
1N4944				440		V dc
1N4946				660		V dc
1N4947				880		V dc
1N4948				1100		V dc
Subgroup 3		T_A = +150°C				
High temperature operation						
Reverse current leakage	4016	V_R = rated V_{RWM} (see 1.3) pulsed (see 4.5.1)	I_{R2}		200	µA
Low temperature operation		T_A = -65°C				
Forward voltage	4011	I_{FM} = 1 A dc pulsed (see 4.5.1)	V_{FM2}	0.6	1.5	V dc
Reverse current	4016	V_R = rated V_{RWM} (see 1.3); dc method	I_{R3}		1.0	µA dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Forward recovery voltage	4026	IF = 0.25 A; t_p = 20 ns (minimum); $t_r \leq$ 8 ns	V_{fr}		5.0	V
Capacitance	4001	V_R = 12 V dc f = 0.1 to 1 MHz	C_J			
1N4942					45	pF
1N4944					35	pF
1N4946					25	pF
1N4947					20	pF
1N4948					15	pF
Reverse recovery time	4031	Condition B1	t_{rr1}			
1N4942					150	ns
1N4944					150	ns
1N4946					250	ns
1N4947					250	ns
1N4948					500	ns
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Group E inspection for (all quality levels) for qualification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Thermal shock (glass strain)	1056	500 cycles, condition A	
Hermetic seal Gross leak	1071		
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 2			22 devices c = 0
Blocking life	1038	80 to 85% of rated V_R, T_A = 150°C, t = 1,000 hours minimum	
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance	3101 or 4081	$R_{\theta JL}$ = 38°C/W maximum; L = .375 inch (9.52)	
Subgroup 5			22 devices c = 0
Barometric pressure (reduced)	1001	1N4944, 1N4946 = 8 mm Hg (100,000 ft) 1N4947, 1N4948 = 33 mmHg (70,000 ft) Voltage during test = V_{RWM}; maximum leakage (I_R) during test shall be 5 μA.	
Subgroups 6 and 7			
Not applicable			

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5. PACKAGING

5.1 Packaging requirement. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

b. Lead finish as specified (see 3.3.1).

c. Type designator and quality assurance level.

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's PIN. This information in no way implies that manufacturers' Part or Identifying Numbers are suitable as substitutes for the military PIN's.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 70, 85
DLA - ES

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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